

HLA regions from 754 haplotypes of five pangenome projects: extraction, typing and pangenome views

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One CWL workflow: pgr-tk extraction, Immuannot typing, pgr-tk and pggb graphs

- ▶ 754 haplotypes: 610 assembled (APR, HPRC release 2, JaSaPaGe) and 144 read from pangenome graphs (K-PanRef, CPC), plus GRCh38 and CHM13
- ▶ MHC (GRCh38 chr6:28.5–33.5 Mb, 100 kb flank) pulled from each haplotype with pgr-tk pgr-query, about 30 s per genome
- ▶ Immuannot (IPD-IMGT/HLA 3.55) gives gene structure and full-resolution allele calls per haplotype
- ▶ Per gene (19 genes): sequences cut at the Immuannot coordinates, pgr-tk principal-bundle plots, pggb graphs rendered with odgi
- ▶ Whole MHC and the class II block (DRA to DMA): pgr-tk bundle decomposition, dendrograms, bundle graphs
- ▶ Second typing method (FuFiHLA) on every individual; T1K on 1000 Genomes Illumina reads
- ▶ Runs on the NIG supercomputer without containers (Toil, `--no-container`); repo github.com/leechuck/pangenome-bh26

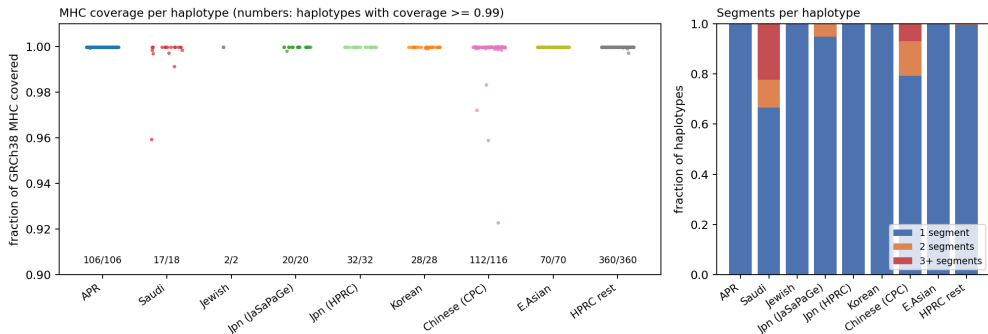
Nine cohorts from five projects; HPRC is split by population and two cohorts exist only as graphs

- ▶ Assemblies
 - ▶ APR, UAE Arab Pangenome Reference: 106 haplotypes
 - ▶ JaSaPaGe: Saudi (ksa*) 18, Japanese (1000G JPT) 20
 - ▶ HPRC release 2, split with the 1000G population labels and the HPRC sample table: Japanese (JPT) 32, Ashkenazi Jewish 2 (HG002 is the only Jewish individual), other East Asian (CHB, CHS, CDX, KHV, HG005) 70, rest 360
- ▶ Pangenome graphs without released assembly files
 - ▶ K-PanRef, Korean pangenome (Minigraph-Cactus): 14 individuals, 28 haplotypes
 - ▶ CPC Phase 1, Chinese Pangenome Consortium (Minigraph-Cactus): 58 individuals, 116 haplotypes; its 3 HPRC samples are not added twice
- ▶ Five JPT individuals were assembled by both HPRC and JaSaPaGe and are kept as separate haplotypes

How: population labels and MHC haplotypes from the Korean and Chinese graphs

- ▶ HPRC split: population code per sample from the 1000G high-coverage sequence indexes (POPULATION column), HPRC release 2 sample metadata for non-1000G samples (HG002 Ashkenazi, HG005 Han Chinese); table `data/hprc_r2_populations.tsv`, used by `make_inputs.py --populations`
- ▶ K-PanRef (`KPanRef.gbz`): all haplotype paths of a sample with `vg paths -F`; only KOREF1 has chromosome-named contigs, so path segments are kept when a minimap2 `asm20` alignment to the GRCh38 MHC has ≥ 50 kb matching bases and `MAPQ  $\geq 20$` (`nig/kpanref_mhc.sbatch`)
- ▶ CPC: the `odgi` file holds only reference paths, so haplotype walks (W lines) are read from the GFA; reference MHC nodes from `odgi extract` of CHM13v2 chr6:28–34 Mb, each walk kept from its first to last MHC node, segment sequences from a second GFA pass, same alignment filter (`nig/cpc_mhc.sbatch`)
- ▶ Both then enter the same extraction and `Immuannot` steps as the assemblies (`nig/add_graph_cohorts.sbatch`)

The MHC comes out complete from 747 of 752 haplotypes, including the graph-derived ones



- ▶ Left: fraction of the GRCh38 MHC covered per haplotype (numbers: haplotypes with ≥ 0.99). Right: MHC segments per haplotype; 720 of 752 on one segment
- ▶ Below 0.99: 4 CPC Chinese haplotypes (lowest 0.92) and 1 JaSaPaGe Saudi (0.96)

How: MHC extraction with pgr-tk (workflow steps prepare_reference, extract_pgrtk)

- ▶ Query: GRCh38 chr6:28,510,120–33,480,577 plus 100 kb flank, cut from the PanSN GRCh38 fasta with samtools faidx
- ▶ Per haplotype: `pgr-query --fastx-file -w 80 -k 56 -r 4 --min-span 64 --merge-range-tol 100000` (SHIMMER index of the input in memory, 8 threads, about 30 s and 9 GB per 3 Gb haplotype)
- ▶ Hits under 20 kb or under 20 anchors dropped; minus-strand hits reverse-complemented; records renamed `sample#hap#contig:start-end`; coverage = query span covered by kept hits (`scripts/pgr_query_rename.py`)
- ▶ Samples present in two projects get a `.cohort` suffix; figure: `analysis/fig2_extraction.py`
- ▶ The five JPT individuals assembled twice: 59 of 61 full-resolution gene calls agree between their HPRC and JaSaPaGe assemblies; the two exceptions differ only in an intron

Allele frequencies separate the cohorts; DRB1 full-length sequences are mostly new

HLA allele landscape by cohort (HPRC r2 split by population; HPRC Jewish = HG002 only) across 752 assembled haplotypes (Immunot on assembly-derived MHC)

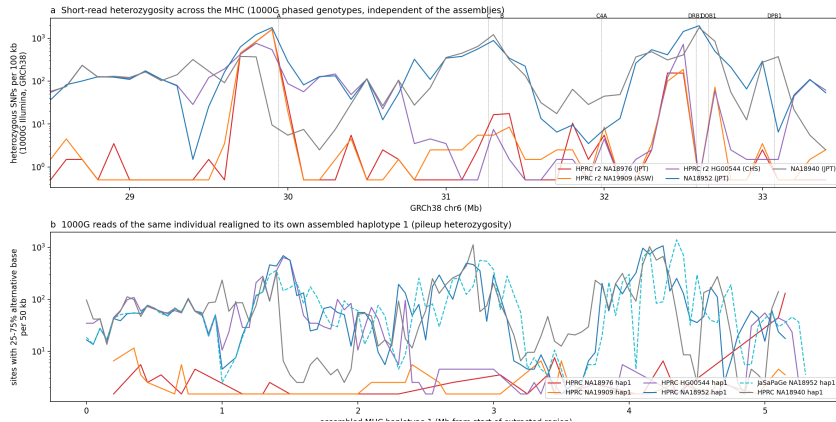


- ▶ A*24:02 in 35–41% of Japanese haplotypes, A*11:01 in 29% of CPC Chinese, B*51:01 in 28% of Saudi, DRB1*03:01 in 18% of APR; DRB1 copies without a full-length database match: 57–79% per cohort (Jewish cohort: HG002 alone)

How: allele calling and aggregation (steps immuannot, aggregate)

- ▶ Immuannot v3 (`immuannot.sh -c <mhc.fa> -r Data-2024Feb02 --diff 0.03, 4 threads`) on each extracted MHC: IPD-IMGT/HLA 3.55 genomic sequences, IPD-KIR 2.13, RefSeq C4; 754 runs
- ▶ Allele = Immuannot consensus field of the transcript row; `:new` marks a full-length sequence absent from the database; novel coding vs noncoding from `cds_distance`
- ▶ `scripts/aggregate_immuannot.py`: `hla_calls.tsv` (one row per gene copy), haplotype x gene matrix, copy numbers (C4A/B long/short, DRB3/4/5 presence), per-cohort two-field frequencies and plots
- ▶ Heat maps: top 6 two-field alleles per cohort, frequency = fraction of haplotypes; figure: `analysis/fig5_population.py`, cohort order and colors in `analysis/cohorts.py`

Two HPRC individuals are truly MHC-homozygous; one JaSaPaGe assembly carries the same haplotype twice



- NA18976, NA19909 (red, orange): under 100 heterozygous 1000G SNPs across 5 Mb, matching their two near-identical assembled haplotypes; JaSaPaGe NA18952 (dashed): own reads show full heterozygosity on both “haplotypes”, hap1 = hap2

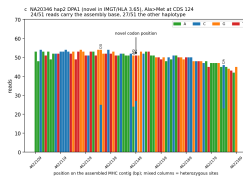
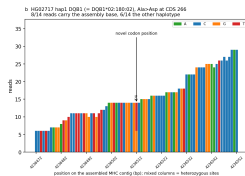
How: assembly-independent and read-level checks (scripts/read_support.sh, analysis/fig3)

- ▶ Candidates: individuals whose two haplotypes carry identical alleles at all eight classical genes (NA18976, NA19909, JaSaPaGe NA18952; HG00544 at seven)
- ▶ Panel a: 1000G high-coverage phased VCF (chr6, 3,202 samples), heterozygous SNPs per 100 kb for the candidates and controls (bcftools view -r chr6:28.51-33.48M)
- ▶ Panel b: MHC-region reads (chr6:28.4–33.6 Mb) from the public 1000G CRAMs, realigned to the individual's own assembled haplotype 1 with minimap2 -ax sr; samtools mpileup, sites with 25–75% alternative base at depth ≥ 15 , counted per 50 kb
- ▶ Panel c: minimap2 -cx asm20 -cs of haplotype 2 against haplotype 1 of the same individual; substitutions per 50 kb from the cs tag
- ▶ JaSaPaGe NA18952 hap2 file carries the HPRC contig name JBHIJT010000011.1; hap1 (hifiasm contig) has the same alleles: the assembly holds one haplotype twice

Of two read-supported novel coding alleles, only NA20346 DPA1 is still novel in IPD-IMGT/HLA 3.65

a Candidate novel coding alleles (Immuannot 'new' with CDS differences) in classical HLA genes. private = variant 31-mer absent from the other 609 haplotypes of the 610-haplotype run; red = differences include indels (typical HiFi homopolymer assembly errors, treat as artefacts); green = heterozygous support in the individual's own short reads

haplotype	cohort	gene	closest known allele	coding difference(s) (known > assembly)	private variants	also on other hap	own 1000G Illumina reads (realigned to assembly)
apr003#1	APR	A	A*24:02:102:01	4 subst.: Val>Leu, Phe>Tyr, Ser>Pro, Lys>Asn	1	no	no public reads
apr011#1	APR	C	C*03:02:02:01	3 subst.: Ile>Leu, Ala>Gly, Gly>Cys	0	no	no public reads
apr001#1	APR	DRB1	DRB1*16:01:01:01	5 subst.: Gln>His, Val>Val, Leu>Arg, Met>Val, Arg>Arg	0	no	no public reads
apr038#1	APR	DRB1	DRB1*14:06:01	5 subst. + indels: Arg>Lys, Val>Ala, Arg>Arg, Cys>Cys, His>Gln	1	no	no public reads
apr002#1	APR	DQB1	DQB1*02:02:01:01	1 subst.: Leu>Leu	0	yes	no public reads
apr015#2	APR	DPB1	DPB1*10:01:01:01	1 subst.: Leu>Leu	1	no	no public reads
HG02717#1	HPRC-Rest (GWD)	DQB1	DQB1*02:02:01:01	Ala>Asp; = DQB1*02:180:02 (IMGT/HLA 3.56): not novel	0	no	8/14 assembly base, 6 other hap
NA20346#2	HPRC-Rest (ASW)	DPA1	DPA1*01:03:01:18Q	Ala>Met; novel in 3.65 (closest 03:02:02, 1 subst.)	1	no	24/51 assembly base, 27 other hap
ksa004#1	JaSaPaGe-Saudi	A	A*02:01:01:134Q	3 indels only (1-bp deletion, 2 insertions)	-	-	no public reads
HG03516#2	HPRC-Rest (ESN)	DPA1	DPA1*03:01:01:01	1-bp insertion only	-	-	-



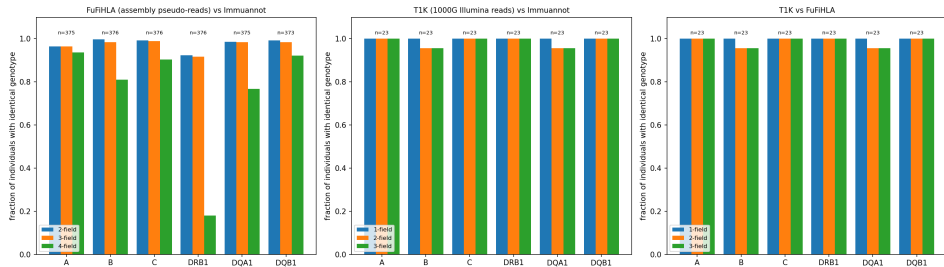
Novel = absent from IPD-IMGT/HLA 3.55 (Immuannot). HG02717 DQB1 equals DQB1*02:180:02 in exons 2–6, created in release 3.56 (Feb 2024).

Pileups: the individual's 1000G Illumina reads realigned to its assembled contig. APR and JaSaPaGe reads are not public

How: novel-allele support (scripts/novel_alleles.py, analysis/fig4)

- ▶ Imm annot cds_mut gives the closest allele and the CDS-relative differences (assembly base first); converted to contig coordinates through the CDS rows, strand-aware
- ▶ Independent-assembly support (610-haplotype run): the 31-mer centred on each variant searched in the same gene cut from all other haplotypes; carriers counted, including the other haplotype of the same individual
- ▶ Read support (HPRC individuals only): 1000G Illumina reads of the same person realigned to the assembled contig (minimap2 -ax sr), samtools mpileup ± 40 bp around the codon
- ▶ Newer database check: coding sequence of each assembled gene aligned with minimap2 -x splice:hq to all coding alleles of IPD-IMGT/HLA 3.65 (T1K index); HG02717: exact over exons 2–6 to DQB1*02:180:02; NA20346: closest DPA1*03:02:02, 1 substitution
- ▶ Indel-only “new” alleles (ksa004 HLA-A, HG03516 DPA1) flagged as likely HiFi homopolymer errors

Assembly-based typing agrees with published and truth-set genotypes at 2-field resolution

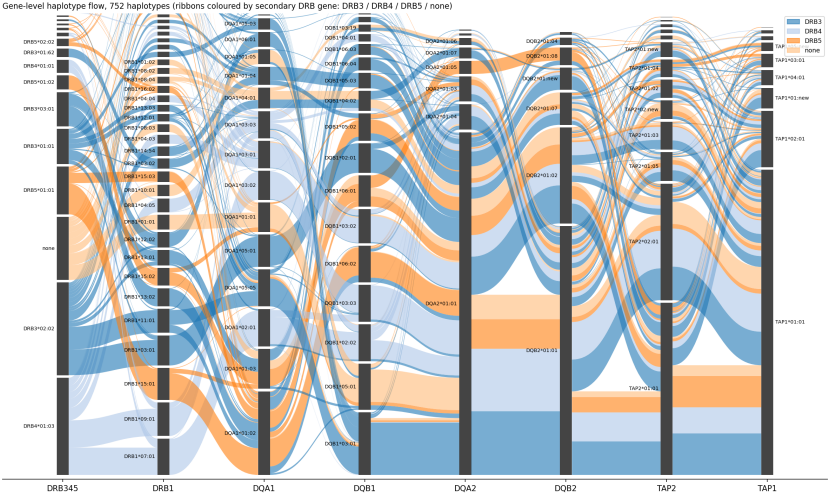


- ▶ Left: FuFiHLA vs Immuannot on the same assemblies, 373–376 individuals; 2-field 92% (DRB1) to 100%. Most DRB1 disagreements: both call a novel allele (19) or FuFiHLA invents a second allele on a homozygous gene (10)
- ▶ Against truth sets: HPRC 4-field labels (Lai et al. 2024, 44 individuals) 2-field Immuannot $\geq 98\%$, FuFiHLA 100%; T1K vs Sanger typing of 412 1000G samples (Gourraud et al. 2014) $\geq 98.8\%$. Middle and right panels: only 23 assembled individuals typed by T1K so far

How: three typing methods and two truth sets (analysis/typing_concordance.py)

- ▶ FuFiHLA 0.2.4 (Bioinformatics 2026, btag231; long-read typing of A, B, C, DRB1, DQA1, DQB1; IPD-IMGT/HLA 3.65): no public long reads for most cohorts, so both assembled MHC haplotypes of an individual are tiled into error-free 15 kb pseudo-reads every 1.5 kb and typed together (nig/fufihla_assemblies.sbatch); checked on HG002, 12 of 12 alleles
- ▶ FuFiHLA artefact test: each reconstructed allele aligned to the individual's assembled gene copies; a consensus absent from the assembly is flagged
- ▶ T1K (Genome Research 2023) on all reads aligned to the MHC, chr6 alternative haplotypes and HLA contigs, extracted from the 2,504 1000G high-coverage CRAMs on the NIG shared data area; alleles with quality 0 ignored (nig/kg_mhc_reads.sbatch)
- ▶ Comparison: unordered allele pairs truncated to 1–4 fields; published ambiguity lists count as a match if any member matches; truth sets in 1000g_ground_truth/ (repository)

518 distinct gene-level class II haplotypes among 752; secondary DRB gene structures the flow

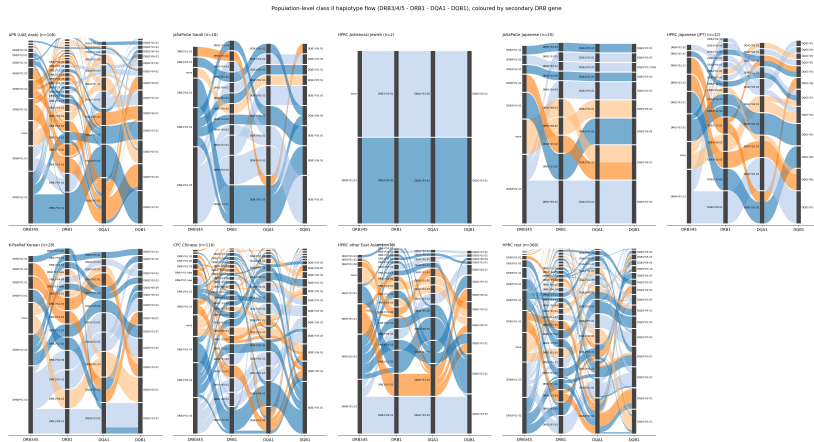


Two-field alleles per haplotype; ribbons colored by DRB3 / DRB4 / DRB5 / none. After Chin, ASHI 2023 (94 haplotypes)

How: class II gene-level haplotype strings and flow plots (analysis/classII_flow_pca.py)

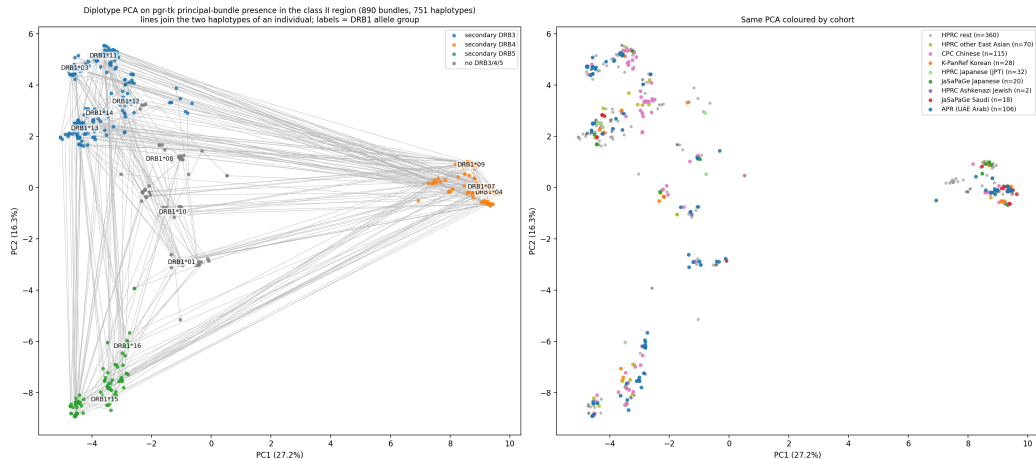
- ▶ Per haplotype: two-field Immuannot allele of DRA, DRB3/4/5 (collapsed into one column, “none” if absent), DRB1, DQA1, DQB1, DQA2, DQB2, DOB, TAP2, TAP1, DMB, DMA, joined into one string
(tables/classII_haplotype_strings.tsv)
- ▶ Alluvial plot: columns are genes, node height = haplotypes carrying the allele, ribbons = haplotypes shared between adjacent alleles, colored by secondary DRB gene
- ▶ Per-cohort version uses the same code on the DRB3/4/5–DRB1–DQA1–DQB1 columns of each cohort
- ▶ Diplotype PCA: one-hot encoding of the six DR/DQ alleles, centered, SVD; both haplotypes of an individual joined by a line

The DR-DQ block differs by population



- DRB4 haplotypes (DR4, DR7, DR9): 44–45% of Saudi and JaSaPaGe Japanese, 39% of Korean; DRB3: 46% of CPC Chinese; DRB1*12:02 16% in CPC Chinese

Sequence-level structure of the class II region follows the DR haplogroups, not the cohort



PCA on presence of 890 pgr-tk principal bundles (DRA to DMA) in 751 haplotypes; PC1 27%, PC2 16%; left: secondary DRB gene, right: cohort;

grey lines join the two haplotypes of an individual

How: class II bundle decomposition with pgr-tk (scripts/extract_span.py, tools/pgr_pbundle.cwl, analysis/classII_bundles.py)

- ▶ Span: from HLA-DRA to HLA-DMA (outermost Immuannot coordinates, 20 kb flank) cut from every MHC haplotype in DRA orientation with samtools faidx; 753 sequences
- ▶ `pgr-pbundle-decomp -w 48 -k 56 -r 2 --min-span 8 --bundle-length-cutoff 500 --bundle-merge-distance 2000`: 890 principal bundles; `pgr-pbundle-bed2dist` for distances and dendrogram, `bed2sorted` for bundle-presence vectors, `bed2svg` for the bundle plot
- ▶ PCA: haplotype x bundle presence matrix, centered, SVD; colors from Immuannot (secondary DRB gene) and cohort labels
- ▶ Dendrogram: Newick from `bed2dist`, leaves mapped to haplotypes, color strips from the allele calls and cohorts; cluster purity at 10–80 clusters in `classII_cluster_purity.tsv`
- ▶ Bundle graph: nodes = bundles from the bed file, edges = consecutive bundles along each haplotype weighted by haplotype count, Kamada-Kawai layout

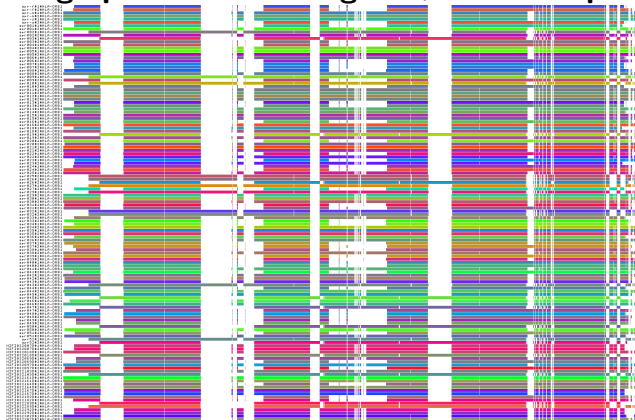
The class II bundle graph is linear except for the DRB block

pngtk principal-bundle graph of the class II region (DRA, DMA), 753 haplotypes, 890 bundles, 1469 adjacencies
node size = bundle length, colour = number of haplotypes; edge width = haplotypes sharing the adjacency



890 principal bundles, node size = bundle length, color = haplotypes carrying it, edges = consecutive bundles on a haplotype

Per-gene pangenome graphs for 19 HLA genes, all 754 haplotypes



- ▶ HLA-DRB1 pggb graph (odgi viz, top rows): rows are haplotypes, x is the graph, color = path coverage
- ▶ Also per gene: pgr-tk bundle plot (SVG/HTML), dendrogram, MAP-graph GFA; whole-MHC bundle plot

How: per-gene sequence sets and graphs (steps `gene_extract`, `gene_bundle`, `gene_graph`, `gene_viz`, `gene_draw`)

- ▶ Gene sequences: every annotated copy cut at its Immuannot coordinates with 2 kb flank, gene orientation, named `sample#hap#GENE` (`scripts/extract_genes.py`)
- ▶ pgr-tk per gene: `pgr-pbundle-decomp -w 48 -k 56 -r 2 --bundle-length-cutoff 200 --bundle-merge-distance 1000`, then `bed2dist` / `bed2sorted` / `bed2svg`
- ▶ pggg 0.7.4 per gene: `-p 90 -s 2000 -n <sequences> -P asm20`; `odgi viz` (1D, rows = haplotypes) and `odgi layout + draw` (2D)
- ▶ Whole MHC: `pgr-pbundle-decomp w=80, k=56, r=6, min_span 28` on the MHC fasta of all haplotypes
- ▶ Runtime of the 754-haplotype aggregation, gene cuts, bundles and graphs: 4.3 h on 16 cores (Toil single machine)

Shared resources for the next steps: 1000G MHC reads and a whole-MHC pangenome graph

- ▶ `/home/asianhla/data/upload/1000G_MHC/`: for each of the 2,504 1000G high-coverage samples, all reads aligned to the MHC, the chr6 alternative haplotypes and the 525 HLA allele contigs (CRAM, about 70 MB per sample), with T1K genotypes; 675 samples done, the rest queued
- ▶ `~leechuck/hla/mhcgraph/out/`: Minigraph-Cactus graph of all 754 MHC haplotypes (GRCh38 reference, CHM13 second reference) with vg giraffe indexes, for aligning the 1000G reads; build running
- ▶ `/home/asianhla/data/upload/HLA/`: results, MHC fastas, Immuannot GTFs, class II, FuFiHLA calls, figures, tables, workflow copy, README

How: 1000G read extraction and graph construction (nig/kg_mhc_reads.sbatch, nig/mhc_mc_graph.sbatch)

- ▶ Reads: `samtools view -C -T GRCh38_full_analysis_set_plus_decoy_hla.fa` on each CRAM for `chr6:28,510,120-33,480,577`, all `chr6*_alt` contigs and all `HLA-*` contigs; primary pairs to fastq; `run-t1k --preset hla` with an IPD-IMGT/HLA index built by `t1k-build.pl`
- ▶ Graph: Cactus 3.3.0 `cactus-pangenome` on the 754 MHC fastas (contigs renamed `ctg<N>`, mapping in `contig_names.tsv`), `--reference GRCh38 CHM13 --gbz clip full --giraffe clip --vcf`; 32 cores, 240 GB
- ▶ The graph covers only the MHC: reads from paralogous sequence elsewhere in the genome can be forced onto it

Take-home

- ▶ Assembly-based HLA typing across five pangenome projects reproduces known population frequencies and reveals data problems (duplicated haplotype, homozygous individuals) that short reads alone would not
- ▶ Pangenome graphs without released assemblies (K-PanRef, CPC) still yield complete MHC haplotypes and allele calls
- ▶ 57–79% of DRB1 gene copies have no full-length match in IPD-IMGT/HLA 3.55; “novel” calls must be rechecked against the newest release (HG02717 DQB1 was named in 3.56)
- ▶ Two assembly-based typing methods agree with truth sets at 2-field resolution; FuFiHLA on assembly pseudo-reads needs a homozygosity check
- ▶ Open: aligning the 1000G MHC reads to the whole-MHC graph; APR and JaSaPaGe raw reads for novel-allele validation; KIR and IGH with the same workflow
- ▶ Workflow, scripts, figures and tables: github.com/leechuck/pangenome-bh26 (h1a/)